

Assignment_2 Part_1

Hypothesis Testing

```
In [1]: import statistics
import numpy as np
from scipy import stats
import pandas as pd
pd.options.display.float_format = '{:,.4f}'.format

In [2]: # define def function for check_normality

def check_normality(data):
    test_stat_normality, p_value_normality=stats.shapiro(data)
    print('p_value={:.4f} & p_value_normality={:.4f}'.format(p_value_normality, p_value_normality))
    if p_value_normality < 0.05:
        print("Reject null hypothesis >> The data is not normally distributed")
    else:
        print("Fail to reject null hypothesis >> The data is normally distributed")

In [3]: # define def function for check_variance_homogeneity

def check_variance_homogeneity(group1, group2):
    test_stat_var, p_value_var=stats.levene(group1, group2)
    print('p_value={:.4f} & p_value_var={:.4f}'.format(p_value_var, p_value_var))
    if p_value_var < 0.05:
        print("Reject null hypothesis >> The variances of the samples are different.")
    else:
        print("Fail to reject null hypothesis >> The variances of the samples are same.")
```

Load Dataset

```
In [4]: fact_table = pd.read_excel("../dataset/sales_data.xlsx", sheet_name = "Fact_table", engine = "openpyxl")
trans_dim = pd.read_excel("../dataset/sales_data.xlsx", sheet_name = "Trans_dim", engine = "openpyxl")
item_dim = pd.read_excel("../dataset/sales_data.xlsx", sheet_name = "item_dim", engine = "openpyxl")
customer_dim = pd.read_excel("../dataset/sales_data.xlsx", sheet_name = "customer_dim", engine = "openpyxl")
time_dim = pd.read_excel("../dataset/sales_data.xlsx", sheet_name = "time_dim", engine = "openpyxl")
store_dim = pd.read_excel("../dataset/sales_data.xlsx", sheet_name = "Store_dim", engine = "openpyxl")

print("Dataset Load Successfully")

Dataset Load Successfully
```

Question 2

The mean sales of item type XX in upazila AA, BB and CC are not the same. (You can calculate the average sales on each week of a month for each upazila. So you should have 12*4=48 data points for each upazila, as there are 4 weeks in a month.) If the means aren't the same, find out which upazila is different using posthoc test.

Merge related Table from Dataset

```
In [6]: # Merge related Table from Dataset ( fact_table, item_dim and time_dim)
fact_item_time = pd.merge(pd.merge(fact_table, item_dim, on = "item_key"), time_dim, on = "time_key")
fact_item_time_customer_dim = pd.merge(fact_item_time, customer_dim, on = "customer_key")
fact_item_time_customer_store_dim = pd.merge(fact_item_time_customer_dim, store_dim, on = "store_key")
fact_item_time_customer_store_dim.head(2)

Out[6]:
```

payment_key	customer_key	time_key	item_key	store_key	quantity_sold	unit_x	unit_price_x	total_price	item_name	...	street	upazila_x	district_x	division_x	store_size	location	city	upazila_y	district_y	division_y
0	P017	C002961	T04782	I00196	S0016	6	ct	16.0000	96.0000	Fiber One 90 Calorie Fudge Brownies	Bishwanath	Bishwanath	Sylhet	Sylhet	medium	Pangsha bridge	Pirojpur	Kawkhali	Pirojpur	Barishal
1	P014	C000195	T00032	I00103	S0020	1	tubs	15.0000	15.0000	Hazelnut International Delight	Lakhan khol	Megna	Comilla	Chittagong	small	687, Adabor	Nilphaman	Sayedpur	Nilphaman	Rangpur

2 rows × 21 columns

Shorting item type "Food - Chips"

```
In [7]: # Filter data for 'Food - Chips' item_type
Food_Chips = fact_item_time_customer_store_dim.query("item_type == 'Food - Chips'")
Food_Chips.head(2)

Out[7]:
```

payment_key	customer_key	time_key	item_key	store_key	quantity_sold	unit_x	unit_price_x	total_price	item_name	...	street	upazila_x	district_x	division_x	store_size	location	city	upazila_y	district_y	division_y
34	P023	C006632	T01374	I00163	S0032	11	bags	15.0000	165.0000	Goldfish Peppardge Cheddar 1.5oz	5/4-A, R62, Kallyerpur	Mirpur-1	Dhaka	Dhaka	medium	House-9, Nawara Garden, Kallyerpur, Mirpur, D..	Dhaka	Mirpur-1	Dhaka	Dhaka
62	P020	C003032	T00815	I00162	S0017	11	Bags	16.0000	176.0000	Gardenof Snacks	514, Iqbal Road	Mohammadpur	Dhaka	Dhaka	large	Rural Development Officer, BRDB Gourmadi, Bar...	Barishal	Gourmadi	Barishal	Barishal

2 rows × 21 columns

Showing filter data

```
In [8]: Food_Chips["item_type"]

Out[8]:
```

	item_type
34	Food - Chips
62	Food - Chips
78	Food - Chips
110	Food - Chips
131	Food - Chips
...	...
99317	Food - Chips
99328	Food - Chips
99945	Food - Chips
99968	Food - Chips
99980	Food - Chips

Name: item_type, length: 4198, dtype: object

Calculate mean sales by upazila,month and week

```
In [33]: # Calculate mean sales by upazila and week
avg_sales_upazila_x = Food_Chips.groupby(["month", "week", "upazila_x", "item_type"]).agg({'total_price': 'mean'}).reset_index()
avg_sales_upazila_y = Food_Chips.groupby(["month", "week", "upazila_y", "item_type"]).agg({'total_price': 'mean'}).reset_index()
# convert total_price to average price
avg_sales_upazila_x.rename(columns={'total_price': 'average_price'}, inplace=True)
avg_sales_upazila_x

In [34]: avg_sales_upazila_x

Out[34]:
```

	month	week	upazila_x	item_type	total_price
0	1	1st Week	Agargaon	Food - Chips	108.0000
1	1	1st Week	Bashundhara R/A	Food - Chips	75.8333
2	1	1st Week	Bishwanath	Food - Chips	76.5000
3	1	1st Week	Chadpur	Food - Chips	118.6667
4	1	1st Week	Dhanmondi	Food - Chips	72.0000
...	...	...	...	...	...
971	12	4th Week	Mohammadpur	Food - Chips	104.7500
972	12	4th Week	Pallabi	Food - Chips	106.6667
973	12	4th Week	Savar	Food - Chips	124.6000
974	12	4th Week	Shyamoli	Food - Chips	133.0000
975	12	4th Week	Uttara	Food - Chips	107.6667

976 rows × 5 columns

```
In [35]: avg_sales_upazila_y

Out[35]:
```

	month	week	upazila_y	item_type	total_price
0	1	1st Week	Balatangi	Food - Chips	165.0000
1	1	1st Week	Bashundhara R/A	Food - Chips	139.0000
2	1	1st Week	Bholahal	Food - Chips	127.5000
3	1	1st Week	Birganj	Food - Chips	96.3333
4	1	1st Week	Bishwanath	Food - Chips	34.0000
...	...	...	...	...	...
1700	12	4th Week	Rangati	Food - Chips	59.5000
1701	12	4th Week	Savar	Food - Chips	53.5000
1702	12	4th Week	Sherpur	Food - Chips	105.5000
1703	12	4th Week	Tahirpur	Food - Chips	99.8000
1704	12	4th Week	Uttara	Food - Chips	59.5000

1705 rows × 5 columns

Extract avarege sales data for each upazila

```
In [36]: # Extract Average sales data for each upazila_y
ea_narsingdi_sadar_avg_sales = avg_sales_upazila_y[avg_sales_upazila_y['upazila_y'] == 'Narsingdi Sadar']
ea_narsingdi_sadar_avg_sales

Out[36]:
```

	month	week	upazila_y	item_type	total_price
27	1	1st Week	Narsingdi Sadar	Food - Chips	150.0000
168	2	1st Week	Narsingdi Sadar	Food - Chips	32.0000
199	2	2nd Week	Narsingdi Sadar	Food - Chips	17.0000
232	2	3rd Week	Narsingdi Sadar	Food - Chips	34.0000
341	3	2nd Week	Narsingdi Sadar	Food - Chips	115.0000
379	3	3rd Week	Narsingdi Sadar	Food - Chips	32.0000
449	4	1st Week	Narsingdi Sadar	Food - Chips	83.0000
480	4	2nd Week	Narsingdi Sadar	Food - Chips	39.5000
515	4	3rd Week	Narsingdi Sadar	Food - Chips	17.0000
555	4	4th Week	Narsingdi Sadar	Food - Chips	64.0000
622	5	2nd Week	Narsingdi Sadar	Food - Chips	153.0000
657	5	3rd Week	Narsingdi Sadar	Food - Chips	40.5000
694	5	4th Week	Narsingdi Sadar	Food - Chips	148.5000
731	6	1st Week	Narsingdi Sadar	Food - Chips	119.3333
768	6	2nd Week	Narsingdi Sadar	Food - Chips	127.0000
839	6	4th Week	Narsingdi Sadar	Food - Chips	116.5000
908	7	2nd Week	Narsingdi Sadar	Food - Chips	17.0000
978	7	4th Week	Narsingdi Sadar	Food - Chips	103.0000
1047	8	2nd Week	Narsingdi Sadar	Food - Chips	80.0000
1082	8	3rd Week	Narsingdi Sadar	Food - Chips	107.6667
1118	8	4th Week	Narsingdi Sadar	Food - Chips	99.0000
1156	9	1st Week	Narsingdi Sadar	Food - Chips	57.3333
1190	9	2nd Week	Narsingdi Sadar	Food - Chips	160.0000
1228	9	3rd Week	Narsingdi Sadar	Food - Chips	102.0000
1262	9	4th Week	Narsingdi Sadar	Food - Chips	64.0000
1300	10	1st Week	Narsingdi Sadar	Food - Chips	198.0000
1335	10	2nd Week	Narsingdi Sadar	Food - Chips	86.0000
1371	10	3rd Week	Narsingdi Sadar	Food - Chips	135.0000
1410	10	4th Week	Narsingdi Sadar	Food - Chips	74.4000
1447	11	1st Week	Narsingdi Sadar	Food - Chips	51.8000
1484	11	2nd Week	Narsingdi Sadar	Food - Chips	102.0000
1520	11	3rd Week	Narsingdi Sadar	Food - Chips	79.0000
1557	11	4th Week	Narsingdi Sadar	Food - Chips	149.0000
1627	12	2nd Week	Narsingdi Sadar	Food - Chips	136.0000
1662	12	3rd Week	Narsingdi Sadar	Food - Chips	152.3333
1698	12	4th Week	Narsingdi Sadar	Food - Chips	76.5000

```
In [37]: # Extract Average sales data for each upazila_x
bb_mirpur_avg_sales = avg_sales_upazila_x[avg_sales_upazila_x['upazila_x'] == 'Mirpur-1']
cc_shyamoli_avg_sales = avg_sales_upazila_x[avg_sales_upazila_x['upazila_x'] == 'Shyamoli']
bb_mirpur_avg_sales

Out[37]:
```

	month	week	upazila_x	item_type	total_price
12	1	1st Week	Mirpur-1	Food - Chips	107.3077
33	1	2nd Week	Mirpur-1	Food - Chips	63.5000
54	1	3rd Week	Mirpur-1	Food - Chips	83.1000
74	1	4th Week	Mirpur-1	Food - Chips	88.6667
94	2	1st Week	Mirpur-1	Food - Chips	99.4000
114	2	2nd Week	Mirpur-1	Food - Chips	101.0769
135	2	3rd Week	Mirpur-1	Food - Chips	87.5000
156	2	4th Week	Mirpur-1	Food - Chips	110.5000
177	3	1st Week	Mirpur-1	Food - Chips	99.7273
195	3	2nd Week	Mirpur-1	Food - Chips	64.7143
215	3	3rd Week	Mirpur-1	Food - Chips	93.8182
235	3	4th Week	Mirpur-1	Food - Chips	86.7000
255	4	1st Week	Mirpur-1	Food - Chips	123.7143
274	4	2nd Week	Mirpur-1	Food - Chips	82.2857
294	4	3rd Week	Mirpur-1	Food - Chips	92.4000
313	4	4th Week	Mirpur-1	Food - Chips	70.0714
334	5	1st Week	Mirpur-1	Food - Chips	109.7778
355	5	2nd Week	Mirpur-1	Food - Chips	86.7692
375	5	3rd Week	Mirpur-1	Food - Chips	62.4000
396	5	4th Week	Mirpur-1	Food - Chips	83.0769
417	6	1st Week	Mirpur-1	Food - Chips	106.1250
437	6	2nd Week	Mirpur-1	Food - Chips	103.6190
456	6	3rd Week	Mirpur-1	Food - Chips	108.6667
477	6	4th Week	Mirpur-1	Food - Chips	115.8000
498	7	1st Week	Mirpur-1	Food - Chips	92.8571
519	7	2nd Week	Mirpur-1	Food - Chips	111.2222
540	7	3rd Week	Mirpur-1	Food - Chips	73.8333
561	7	4th Week	Mirpur-1	Food - Chips	120.0909
581	8	1st Week	Mirpur-1	Food - Chips	112.2500
601	8	2nd Week	Mirpur-1	Food - Chips	118.2222
619	8	3rd Week	Mirpur-1	Food - Chips	102.3636
640	8	4th Week	Mirpur-1	Food - Chips	109.1000
660	9	1st Week	Mirpur-1	Food - Chips	117.0000
681	9	2nd Week	Mirpur-1	Food - Chips	95.6923
702	9	3rd Week	Mirpur-1	Food - Chips	79.3333
722	9	4th Week	Mirpur-1	Food - Chips	77.4706
743	10	1st Week	Mirpur-1	Food - Chips	104.8462
764	10	2nd Week	Mirpur-1	Food - Chips	86.4615
785	10	3rd Week	Mirpur-1	Food - Chips	97.6875
806	10	4th Week	Mirpur-1	Food - Chips	86.2500
825	11	1st Week	Mirpur-1	Food - Chips	113.5000
846	11	2nd Week	Mirpur-1	Food - Chips	95.7143
867	11	3rd Week	Mirpur-1	Food - Chips	109.8182
888	11	4th Week	Mirpur-1	Food - Chips	105.2500
907	12	1st Week	Mirpur-1	Food - Chips	114.1818
927	12	2nd Week	Mirpur-1	Food - Chips	92.1250
948	12	3rd Week	Mirpur-1	Food - Chips	119.3529
968	12	4th Week	Mirpur-1	Food - Chips	99.6667

```
In [38]: cc_shyamoli_avg_sales

Out[38]:
```

	month	week	upazila_x	item_type	total_price
18	1	1st Week	Shyamoli	Food - Chips	89.2500
39	1	2nd Week	Shyamoli	Food - Chips	30.0000
60	1	3rd Week	Shyamoli	Food - Chips	77.8000
80	1	4th Week	Shyamoli	Food - Chips	117.3333
99	2	1st Week	Shyamoli	Food - Chips	73.0000
120	2	2nd Week	Shyamoli	Food - Chips	15.0000
141	2	3rd Week	Shyamoli	Food - Chips	150.0000
162	2	4th Week	Shyamoli	Food - Chips	136.0000
183	3	1st Week	Shyamoli	Food - Chips	74.0000
200	3	2nd Week	Shyamoli	Food - Chips	94.0000
221	3	3rd Week	Shyamoli	Food - Chips	105.0000
241	3	4th Week	Shyamoli	Food - Chips	82.0000
279	4	2nd Week	Shyamoli	Food - Chips	161.5000
319	4	4th Week	Shyamoli	Food - Chips	85.8333
340	5	1st Week	Shyamoli	Food - Chips	60.6667
361	5	2nd Week	Shyamoli	Food - Chips	105.0000
381	5	3rd Week	Shyamoli	Food - Chips	88.2857
402	5	4th Week	Shyamoli	Food - Chips	137.0000
423	6	1st Week	Shyamoli	Food - Chips	17.0000
442	6	2nd Week	Shyamoli	Food - Chips	83.5000
462	6	3rd Week	Shyamoli	Food - Chips	95.2500
483	6	4th Week	Shyamoli	Food - Chips	79.2500
504	7	1st Week	Shyamoli	Food - Chips	170.0000
525	7	2nd Week	Shyamoli	Food - Chips	51.0000
546	7	3rd Week	Shyamoli	Food - Chips	55.3333
567	8	1st Week	Shyamoli	Food - Chips	62.4000
607	8	2nd Week	Shyamoli	Food - Chips	34.0000
625	8	3rd Week	Shyamoli	Food - Chips	23.5000
646	8	4th Week	Shyamoli	Food - Chips	112.8333
666	9	1st Week	Shyamoli	Food - Chips	132.2500
687	9	2nd Week	Shyamoli	Food - Chips	88.6667
708	9	3rd Week	Shyamoli	Food - Chips	56.0000
728	9	4th Week	Shyamoli	Food - Chips	68.0000
749	10	1st Week	Shyamoli	Food - Chips	124.0000
770	10	2nd Week	Shyamoli	Food - Chips	128.0000
791	10	3rd Week	Shyamoli	Food - Chips	131.0000
812	10	4th Week	Shyamoli	Food - Chips	95.0000
831	11	1st Week	Shyamoli	Food - Chips	94.5000
852	11	2nd Week	Shyamoli	Food - Chips	113.8333
873	11	3rd Week	Shyamoli	Food - Chips	103.1429
894	11	4th Week	Shyamoli	Food - Chips	105.0000
933	12	2nd Week	Shyamoli	Food - Chips	130.3333
953	12	3rd Week	Shyamoli	Food - Chips	58.7500
974	12	4th Week	Shyamoli	Food - Chips	133.0000

Unpaired more than 2 groups

Normality Check

```
In [39]: check_normality(ea_narsingdi_sadar_avg_sales.total_price)
check_normality(bb_mirpur_avg_sales.total_price)
check_normality(cc_shyamoli_avg_sales.total_price)

p value:0.3299
Fail to reject null hypothesis >> The data is normally distributed
p value:0.1525
Fail to reject null hypothesis >> The data is normally distributed
p value:0.7910
Fail to reject null hypothesis >> The data is normally distributed
```

Leven Check

```
In [40]: stat, p_value levene = stats.levene(ea_narsingdi_sadar_avg_sales.total_price, bb_mirpur_avg_sales.total_price,
print('p_value={:.4f} & p_value levene={:.4f}'.format(p_value, p_value levene))
if p_value levene < 0.05:
    print("Reject null hypothesis >> The variances of the samples are different.")
else:
    print("Fail to reject null hypothesis >> The variances of the samples are same.")

p value:0.0000
Reject null hypothesis >> The variances of the samples are different.
```

```
In [41]: F, p_value = stats.f_oneway(ea_narsingdi_sadar_avg_sales.total_price, bb_mirpur_avg_sales.total_price,
print('p_value={:.4f} & p_value')

if p_value < 0.05:
    print("Reject null hypothesis")
else:
    print("Fail to reject null hypothesis")

p value:0.69455
Fail to reject null hypothesis
```

```
In [ ]:

In [ ]:
```